

COMPUTER SCIENCE TRIPOS Part IB 75%, Part II 50% – 2021 – Paper 7

2 Economics, Law and Ethics (ah793)

The new Master of (fictional) Porterhouse would like to modernise the college. They bring the following proposals to the College Council, of which you are a member.

- (a) Proposal 1: Change the way students are admitted to the college for all subjects. Rather than having humans interview candidates, or at least read their references and personal statements, an algorithm would make all admission decisions.

- (i) Use an ethical perspective to make a persuasive argument for or against Proposal 1, defining and explaining relevant concepts. [5 marks]

Philosophies of ethics;
contemporary ethical issues

Answer: The argument for algorithmic decision making is that it is more efficient. For example, you don't need to mobilise thousands of students to come to Cambridge to be interviewed in-person, taking up weeks of interviewer and professional service staff time. However, this is an economic rather than an ethical argument in favour of algorithmic decision making (1 mark). Ethically, the concern is that the algorithm is likely to introduce negative outcomes, especially where bias exists (1 mark). It would be difficult to argue that the data that an algorithm is likely to be based on would not be biased (1 mark). Historically, university colleges have been more likely to admit white men from high socio-economic backgrounds and elite public schools (1 mark). Those who have had to work harder to overcome disadvantage would therefore likely to continue to be overlooked. This concern can be viewed through the lens of Rawlsian ethics, in that we should maximise the welfare of the most disadvantaged (1 mark).

- (ii) If Proposal 1 is accepted, what changes would the college need to implement to comply with GDPR? [5 marks]

Law and the Internet

Answer: The GDPR provides individuals with rights in relation to automated decision making and profiling (2 marks). Applicants would need to be informed, such as through a privacy policy (1 mark) about the existence of automated decision making, including profiling and information about how decisions are made, the significance and the consequences (2 marks).

- (b) Proposal 2: The college is buying apartments in the next phase of the North West Cambridge development for use by graduate students and students with dependents. The options are to buy a) a range of apartments, from utilitarian to luxurious, which would be rented out at different amounts; or b) a block of apartments all of the same size, which would be rented out at the same price.

- (i) Use an economic perspective to make a persuasive argument for option a) or b), defining and explaining relevant concepts. [5 marks]

Classical economics and consumer theory

Answer: To capture the consumer surplus (define for 1 mark), the option a) would be a Pareto improvement (define for 1 mark) over option b), in that some people are better off but nobody is worse off (1 mark). The college will be able to generate more rental income from the higher grade apartments, while the students pay for what they can afford (1 mark). This is a form of price discrimination (define for 1 mark) (namely, versioning) .

- (ii) If Proposal 2, option a) is accepted, how might the college decide how much to charge for each grade of apartment? [5 marks]

Answer: The college may like to ascertain students' reservation price (define for 1 mark) for the amenities offered within each grade apartment (2 mark). They can then price each grade of apartment where the supply and demand curves cross (the market equilibrium price) (2 marks).

An alternative answer might be to invite students to tender a rent for the apartments and allocate them using a second price auction. For example, at the start of each academic year students could bid the maximum they are willing to pay for the minimum utility. An answer taking this route would need to define the type of auction they select and how it would take place (2 marks), why they selected that type (2 marks). The answer would need to factor in that each student should end up renting one apartment, no more and no less, e.g. basic apartments may need to be provided at an affordable minimum rate (1 mark).
